

Parametric Equations & Modeling

ANSWER KEY



Evaluating parametric equations

- 1 Given $x(t) = t^2 - 1$ and $y(t) = 2t + 3$, find the point on the curve at $t = 2$.
 $x(2) = 4 - 1 = 3$. $y(2) = 4 + 3 = 7$. Point: $(3, 7)$.
- 2 Given $x(t) = 3\cos t$ and $y(t) = 3\sin t$, find the point at $t = \frac{\pi}{3}$, and identify the shape traced by the curve.
 $x(\frac{\pi}{3}) = 3\cos(\frac{\pi}{3}) = 3(\frac{1}{2}) = \frac{3}{2}$. $y(\frac{\pi}{3}) = 3\sin(\frac{\pi}{3}) = 3(\frac{\sqrt{3}}{2}) = \frac{3\sqrt{3}}{2}$. Since $x^2 + y^2 = 9\cos^2 t + 9\sin^2 t = 9$, the curve traces a circle of radius 3 centered at the origin.

Eliminating the parameter

- 3 Eliminate the parameter from $x(t) = t + 1$, $y(t) = t^2 - 2$ to find the rectangular equation, and identify the curve.
From $x = t + 1$: $t = x - 1$. Substitute: $y = (x - 1)^2 - 2 = x^2 - 2x + 1 - 2 = x^2 - 2x - 1$. This is a parabola.
- 4 Eliminate the parameter from $x(t) = 2t - 3$, $y(t) = 4t + 1$ to find the rectangular equation.
From $x = 2t - 3$: $t = \frac{x + 3}{2}$. Substitute: $y = 4(\frac{x + 3}{2}) + 1 = 2(x + 3) + 1 = 2x + 7$. This is a line: $y = 2x + 7$.

Visualizing a parametric curve's components

- 5 The graph shows the two component functions $x(t) = t^2$ (upper curve) and $y(t) = 2t$ (lower, straight line) for the parametric curve $x(t) = t^2$, $y(t) = 2t$, plotted separately against t on the horizontal axis.
 - a Eliminate the parameter t to find the rectangular equation relating x and y for this curve. From $y = 2t$: $t = \frac{y}{2}$. Substituting into $x = t^2$: $x = (\frac{y}{2})^2 = \frac{y^2}{4}$, so $y^2 = 4x$.
 - b Use your rectangular equation to describe the shape of the actual parametric curve (not the component graphs shown). $y^2 = 4x$ is a sideways parabola opening to the right, with vertex at the origin — very different from the two component graphs shown, which are just x and y plotted individually against the parameter t .

Parametric equations for lines and conics

- 6 Write parametric equations for the line segment from $(1, 2)$ to $(5, 10)$, using $t \in [0, 1]$.
 $x(t) = 1 + 4t$, $y(t) = 2 + 8t$ for $0 \leq t \leq 1$ (since the change in x is $5 - 1 = 4$ and the change in y is $10 - 2 = 8$). Check: at $t = 0$, $(1, 2)$; at $t = 1$, $(5, 10)$ ✓.

- 7 Write parametric equations for an ellipse with semi-major axis 5 (horizontal) and semi-minor axis 3 (vertical), centered at the origin.
- $x(t) = 5\cos t$, $y(t) = 3\sin t$, for $0 \leq t < 2\pi$. Check: $\left(\frac{x}{5}\right)^2 + \left(\frac{y}{3}\right)^2 = \cos^2 t + \sin^2 t = 1$, matching the standard ellipse equation.
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Modeling projectile motion parametrically

- 8 A ball is launched with initial speed 40 m/s at an angle of 30° above horizontal. Its position is modeled by $x(t) = 40\cos(30^\circ)t$ and $y(t) = 40\sin(30^\circ)t - 4.9t^2$. Find the ball's position at $t = 2$ seconds, to one decimal place.
- $x(2) = 40\cos(30^\circ)(2) \approx 40(0.866)(2) \approx 69.3$ m.
 $y(2) = 40\sin(30^\circ)(2) - 4.9(4) = 40(0.5)(2) - 19.6 = 40 - 19.6 = 20.4$ m. Position: approximately (69.3, 20.4).
- 9 For the same projectile, find the time at which the ball lands (returns to $y = 0$).
- Set $y(t) = 0$: $20t - 4.9t^2 = 0$, so $t(20 - 4.9t) = 0$, giving $t = 0$ or $t = \frac{20}{4.9} \approx 4.08$ seconds. The ball lands at $t \approx 4.08$ s.
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Comparing parametric speed and direction

- 10 Two objects follow the same path $y = x^2$, but object A uses $x(t) = t$, $y(t) = t^2$ while object B uses $x(t) = 2t$, $y(t) = 4t^2$. Explain why they trace the same curve but do not move through it at the same rate.
- Both satisfy $y = x^2$ (for B: $y = 4t^2 = (2t)^2 = x^2$ ✓), so they trace identical curves. But object B's x -coordinate changes twice as fast per unit of t as object A's, meaning B moves through the same path twice as fast — the parametrization encodes speed and direction of travel, which the rectangular equation $y = x^2$ alone does not capture.
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Restricting the parameter's domain

- 11 The parametric equations $x(t) = \cos t$, $y(t) = \sin t$ trace the full unit circle for $t \in [0, 2\pi)$. Describe the portion of the circle traced when the domain is restricted to $t \in \left[0, \frac{\pi}{2}\right]$.
- Restricting to $t \in \left[0, \frac{\pi}{2}\right]$ traces only the arc from (1, 0) (at $t = 0$) to (0, 1) (at $t = \frac{\pi}{2}$) — the quarter of the circle lying in Quadrant I.
- 12 Explain why restricting t in $x(t) = t^2$, $y(t) = t$ to $t \geq 0$ changes the rectangular relation from $x = y^2$ (a full sideways parabola) to only its upper half.
- Eliminating the parameter gives $x = y^2$ for all t , but since $y = t$ directly, restricting $t \geq 0$ forces $y \geq 0$ as well — so only the upper half of the sideways parabola (where $y \geq 0$) is actually traced, even though the unrestricted equation $x = y^2$ would include the lower half too.
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Parametric equations for circular motion with a phase shift

- 13** A point moves along a circle of radius 4 according to $x(t) = 4\cos\left(t + \frac{\pi}{4}\right)$, $y(t) = 4\sin\left(t + \frac{\pi}{4}\right)$. Find its position at $t = 0$, and explain the effect of the $\frac{\pi}{4}$ phase shift compared to $x(t) = 4\cos t$, $y(t) = 4\sin t$.
At $t = 0$: $x(0) = 4\cos\left(\frac{\pi}{4}\right) = 4\left(\frac{\sqrt{2}}{2}\right) = 2\sqrt{2}$, $y(0) = 4\sin\left(\frac{\pi}{4}\right) = 2\sqrt{2}$. Without the phase shift, the point would start at $(4, 0)$ when $t = 0$; the added $\frac{\pi}{4}$ advances the starting position by an eighth of a full revolution counterclockwise, to $(2\sqrt{2}, 2\sqrt{2})$, while the circle traced and the speed of travel remain unchanged.